

2-Mark Questions

Q.1 Define Digital Forensics. (Unit 1)

Digital forensics is the process of **identifying, preserving, analyzing, and presenting digital evidence** in a legally acceptable manner. It involves recovering and examining data from electronic devices to investigate cybercrimes or policy violations.

Example: Recovering deleted emails from a suspect's laptop to trace communication.

Q.2 List out steps involved in handling Electronic Media. (Unit 1)

The main steps in handling electronic media are:

- **Identification:** Recognize and label potential digital evidence.
- **Preservation:** Protect the media from alteration or damage.
- **Collection:** Acquire data using forensic tools.
- **Analysis:** Examine data for relevant information.
- **Documentation:** Maintain chain-of-custody records.

Example: Creating a bit-by-bit image of a seized hard drive.

Q.3 Restate the term Cryptography. (Unit 1)

Cryptography is the science of **converting readable data (plaintext) into unreadable form (ciphertext)** to protect it from unauthorized access. It ensures **confidentiality, integrity, and authentication** of data during transmission or storage.

Example: Using AES encryption to secure online transactions.

Q.4 Define Cardinal Rules of Forensics. (Unit 3)

The cardinal rules of forensics are the fundamental principles that ensure the integrity of digital evidence:

1. **Do not alter original evidence.**
2. **Maintain proper chain of custody.**
3. **Document every forensic action.**

Following these rules ensures evidence remains admissible in court.

Example: Using a write blocker to prevent changes to a suspect's hard drive.

Q.5 Define Cyber Law. (Unit 2)

Cyber law refers to the **legal framework that governs activities conducted through computers and the internet**. It addresses issues such as cybercrimes, online fraud, hacking, privacy, and digital rights.

Example: The Information Technology Act, 2000, regulates cybercrimes and electronic transactions in India.

Q.6 List out types of File Systems used in Windows OS. (Unit 3)

The main file systems used in Windows operating systems are:

- **FAT12** – Used in floppy disks and early systems.
 - **FAT16** – Supports larger partitions than FAT12.
 - **FAT32** – Common in older Windows versions and removable drives.
 - **NTFS** – Default for modern Windows; supports security, compression, and large files.
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Q.7 Restate the definition of FAT File System. (Unit 3)

The FAT (File Allocation Table) file system is a simple file management system that uses a table to record the location of files on a disk. It tracks clusters of data, allowing the operating system to read, write, and manage files efficiently.

Example: FAT32 is widely used on USB drives and memory cards.

Q.8 Define Anti-Forensics. (Unit 4)

Anti-forensics refers to techniques used to **conceal, alter, or destroy digital evidence** to hinder forensic investigations. It includes file wiping, data encryption, metadata alteration, and log deletion to make recovery or analysis difficult.

Example: Using a file shredder to erase traces of deleted documents.

Q.9 Define Computer Artifact. (Unit 3)

A computer artifact is any **residual data or trace left by system or user activity** that can be analyzed during an investigation. It helps reconstruct events or user actions.

Example: Windows registry entries, browser history, or event logs are common artifacts.

Q.10 List out UNIX file systems. (Unit 3)

Common UNIX file systems include:

- **UFS (Unix File System)**
- **EXT2 (Second Extended File System)**
- **EXT3 and EXT4 (with journaling features)**
- **ReiserFS and XFS**

These systems manage data through inodes and directories for efficient storage and retrieval.

Q.11 Restate the meaning of OS Artifact. (Unit 3)

An OS artifact is a **piece of residual data created automatically by an operating system** during normal usage. These artifacts help forensic experts reconstruct user activities and system events. They include files, logs, caches, and metadata that indicate how and when the system was used.

Example: Windows registry entries, event logs, and prefetch files are common OS artifacts used in investigations.

Q.12 Define Vulnerability Assessment. (Unit 4)

Vulnerability assessment is a **systematic process of identifying, analyzing, and prioritizing weaknesses** in computer systems, networks, or software applications. The goal is to find and fix vulnerabilities before attackers can exploit them. It's a crucial step in maintaining cyber security and risk management.

Example: Security tools like Nessus or OpenVAS are used to scan systems for potential security flaws.

Q.13 Define Hacking. (Unit 1)

Hacking is the act of **gaining unauthorized access to computer systems, networks, or digital devices** to steal, manipulate, or destroy data. It can be malicious (black-hat) or ethical (white-hat), depending on intent and authorization.

Example: Breaking into a corporate server to steal customer information is considered malicious hacking.

Q.14 List out Security Laws in India. (Unit 2)

Key cyber and security laws in India include:

- **Information Technology (IT) Act, 2000** – Regulates cybercrimes, electronic records, and digital signatures.
 - **Indian Penal Code (IPC), 1860** – Addresses fraud, identity theft, and data tampering.
 - **Copyright Act, 1957** – Protects digital and software intellectual property.
- Together, these laws establish a legal framework for cyber security.
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Q.15 Restate importance of Security Assurance. (Unit 2)

Security assurance is important because it **ensures that information systems operate safely and securely under all conditions**. It involves testing and validating security controls to confirm that data integrity, confidentiality, and availability are maintained. It builds user trust and helps organizations meet compliance standards.

Example: Conducting regular ISO 27001 security audits provides assurance that systems are well-protected.

Q.16 Define Digital Investigation. (Unit 5)

Digital investigation is the **systematic process of collecting, analyzing, and interpreting digital evidence** from computers or electronic devices to uncover facts related to cybercrimes or security incidents. It helps identify perpetrators, reconstruct events, and present findings in a legally admissible form.

Example: Tracing unauthorized access to a company's server through log file analysis.

Q.17 State any two types of Cyber Crimes. (Unit 1)

Two common types of cybercrimes are:

1. **Phishing:** Fraudulent attempts to obtain sensitive information through fake emails or websites.
2. **Identity Theft:** Stealing personal data to impersonate someone for financial or social gain.

Example: Using stolen credit card information to make unauthorized purchases online.

Q.18 Rewrite the definition of Hacking. (Unit 1)

Hacking refers to the **unauthorized intrusion into a computer system, network, or digital device** to access, steal, or manipulate data. It can be ethical (authorized testing) or unethical (criminal activity).

Example: A hacker breaching a company's database to steal customer information is an act of unethical hacking.

Q.19 Write the full form of IPR. (Unit 2)

The full form of **IPR** is **Intellectual Property Rights**. These are **legal rights granted to individuals or organizations** to protect their creative works, inventions, software, or digital content from unauthorized use or duplication.

Example: Copyrights and patents are common types of intellectual property rights.

Q.20 State the use of Disk Imaging. (Unit 4)

Disk imaging is used to **create an exact replica of a digital storage device** for forensic examination. It preserves all data, including deleted and hidden files, without altering the original evidence.

Example: Investigators use tools like FTK Imager to create a forensic image of a suspect's hard drive.

Q.21 State any two International Standards for Information Security. (Unit 2)

Two major international standards for information security are:

1. **ISO/IEC 27001:** Specifies requirements for establishing and maintaining an effective Information Security Management System (ISMS).
2. **ISO/IEC 27002:** Provides guidelines and best practices for implementing security controls.

Example: Organizations use ISO 27001 certification to demonstrate strong data protection practices.

Q.22 Rewrite the meaning of Security Audit. (Unit 2)

A security audit is a **systematic evaluation of an organization's information systems**

to ensure compliance with security policies, standards, and regulations. It identifies vulnerabilities, verifies the effectiveness of controls, and recommends improvements.
Example: Conducting an audit to check firewall configurations and data access permissions for compliance.

Q.23 Write the purpose of Disk Imaging tools. (Unit 4)

Disk imaging tools are used to **create exact bit-by-bit copies of digital storage devices** for forensic or backup purposes. They ensure that evidence is preserved without modifying the original data.

Example: Tools like FTK Imager and EnCase are used to create forensic images of hard drives during investigations.

Q.24 State the role of Cryptography in Forensics. (Unit 1)

Cryptography in forensics is used to **protect, analyze, or decrypt digital evidence** during investigations. It ensures secure storage and transmission of data while also helping forensic experts access encrypted evidence legally.

Example: Investigators use decryption techniques to recover data from an encrypted hard drive.

Q.25 State any two Cyber Security Laws. (Unit 2)

Two key cybersecurity laws are:

1. **Information Technology (IT) Act, 2000** – Regulates cybercrimes, electronic records, and data protection in India.
2. **General Data Protection Regulation (GDPR)** – An international law that governs data privacy and user consent in the European Union.

Example: The IT Act punishes offenses like hacking and identity theft.

Q.26 Rewrite the definition of Security Assurance. (Unit 2)

Security assurance is the **confidence that information systems are protected against potential threats and function securely under all conditions**. It involves verifying that implemented security controls effectively safeguard data confidentiality, integrity, and availability through audits, testing, and continuous monitoring.

Example: Conducting penetration tests to ensure a system's resilience against cyberattacks.

Q.27 Write any two examples of Digital Devices used in Forensics. (Unit 1)

Two common digital devices used in forensic investigations are:

1. **Hard Drives:** Store crucial evidence such as files, logs, and system data.

2. **Mobile Phones:** Contain call records, messages, and GPS data vital for analysis.

Example: Extracting deleted messages from a suspect's smartphone using forensic tools.

Q.28 State the purpose of Authentication in Forensics. (Unit 3)

Authentication in forensics ensures that **digital evidence remains original, untampered, and verifiable** throughout the investigation. It validates that data analyzed is the same as when it was collected, maintaining legal credibility.

Example: Using hash values like MD5 or SHA-1 to verify the integrity of a forensic image.

Q.29 State any two types of Forensic Tools. (Unit 4)

Two main types of forensic tools are:

1. **Disk Imaging Tools:** Used to create exact replicas of storage media (e.g., FTK Imager).
2. **Data Recovery Tools:** Retrieve deleted or lost files from damaged or formatted drives.

Example: EnCase and Autopsy are popular forensic software used for evidence analysis.

Q.30 Rewrite the definition of Data Recovery. (Unit 4)

Data recovery is the **process of restoring lost, deleted, or inaccessible data** from storage devices such as hard drives, USBs, or memory cards. It helps investigators retrieve vital evidence that may have been intentionally erased or accidentally lost.

Example: Recovering deleted documents using forensic tools like Recuva or R-Studio.

Q.31 Write the function of EnCase tool. (Unit 4)

EnCase is a **professional digital forensic tool** used to acquire, analyze, and report on digital evidence from various storage media. It helps investigators recover deleted files, examine file systems, and generate detailed forensic reports while maintaining data integrity.

Example: EnCase can analyze hard drives to locate hidden or deleted documents.

Q.32 State two features of NTFS. (Unit 3)

Two key features of NTFS (New Technology File System) are:

1. **File Security:** Supports access control lists (ACLs) to manage user permissions.
2. **File Compression and Encryption:** Reduces storage usage and enhances data protection.

Example: NTFS allows encrypting a sensitive file using Windows' built-in EFS (Encrypting File System).

Q.33 Identify different sources of Electronic Evidence. (Unit 1)

Different sources of electronic evidence include **computers, mobile devices, servers, cloud storage, and external drives**. Logs, emails, CCTV footage, and social media accounts can also provide crucial evidence.

Example: Email logs from an organization's server can serve as electronic evidence in fraud investigations.

Q.34 Classify various types of Internet Crimes. (Unit 1)

Internet crimes can be classified as:

1. **Financial Crimes** – Online frauds, phishing, and credit card theft.
2. **Cyber Harassment** – Cyberbullying or defamation.
3. **Hacking-related Crimes** – Unauthorized access or data breaches.

Example: Identity theft through fake websites is a common internet crime.

Q.35 Give example of a Security Audit Standard. (Unit 2)

A common security audit standard is **ISO/IEC 27001**, which defines the requirements for implementing and maintaining an **Information Security Management System (ISMS)**. It helps organizations assess, manage, and mitigate information security risks effectively.

Example: A company following ISO 27001 ensures secure handling of customer data and compliance with legal standards.

Q.36 Enlist any two Anti-Forensics techniques. (Unit 4)

Two common anti-forensics techniques are:

1. **Data Wiping:** Permanently erasing files to prevent recovery.
2. **Encryption:** Protecting data using cryptographic keys to make it unreadable without authorization.

These methods are used by attackers to hide evidence or obstruct forensic investigations.

Example: Using software like CCleaner to wipe sensitive files securely.

Q.37 Identify the components of Digital Evidence. (Unit 5)

The key components of digital evidence include:

1. **Data Content:** Actual information stored such as files, emails, or logs.
2. **Metadata:** Information about data, including timestamps and file paths.

3. **Chain of Custody:** Documentation of how evidence is handled.

Example: A document's creation date and author name serve as metadata in evidence.

Q.38 Classify Cyber Crimes based on motive. (Unit 1)

Cybercrimes can be classified based on motive as:

1. **Financial Motive:** Crimes like online fraud, ransomware, and credit card theft.
2. **Political or Social Motive:** Hactivism or cyber-terrorism.
3. **Personal Motive:** Revenge, harassment, or defamation.

Example: Launching ransomware to extort money represents a financially motivated cybercrime.

Q.39 Give example of Forensic Software Tool. (Unit 4)

A popular forensic software tool is **Autopsy**, an open-source platform used for disk imaging, data recovery, and file analysis. It provides graphical interfaces for examining digital evidence and generating reports.

Example: Autopsy helps investigators recover deleted files and trace user activity on a computer system.

Q.40 Enlist any two modules of Digital Investigation. (Unit 5)

Two essential modules of digital investigation are:

1. **Data Acquisition:** Collecting digital evidence from devices in a forensically sound manner.
2. **Analysis and Reporting:** Examining acquired data to extract useful information and document findings.

Example: Using forensic tools to acquire disk images and analyze deleted email traces.

Q.41 Identify any two Internet Crimes. (Unit 1)

Two common Internet crimes are:

1. **Phishing:** Deceptive emails or websites used to steal personal data such as passwords or banking details.
2. **Cyberstalking:** Using the internet to harass or threaten individuals repeatedly.

Example: Creating a fake banking website to trick users into revealing login credentials.

Q.42 Classify Electronic Media types. (Unit 1)

Electronic media can be classified into:

1. **Primary Storage Devices:** Such as hard drives, SSDs, and flash memory.

2. **Secondary or Removable Media:** Includes CDs, DVDs, USB drives, and memory cards.

Both are crucial sources of evidence in forensic investigations.

Example: A forensic expert may analyze a USB drive containing hidden files.

Q.43 Give example of Forensic Investigation Tool. (Unit 4)

A widely used forensic investigation tool is **FTK (Forensic Toolkit)**, which helps investigators acquire, analyze, and visualize digital evidence. It supports file recovery, email analysis, and keyword searching while maintaining data integrity.

Example: FTK can recover deleted files and generate a timeline of user activity.

Q.44 Enlist any two methods to recover data from mobiles. (Unit 5)

Two methods to recover data from mobile devices are:

1. **Logical Extraction:** Retrieves accessible data like messages and contacts using specialized tools.
2. **Physical Extraction:** Acquires complete device memory, including deleted data.

Example: Using Cellebrite UFED to extract call logs and deleted messages from smartphones.

Q.45 Identify any two types of Electronic Evidence. (Unit 1)

Two common types of electronic evidence are:

1. **Direct Evidence:** Includes digital files like emails, images, and documents.
2. **Metadata Evidence:** Provides contextual details such as timestamps and file origins.

Example: An email containing incriminating content and its header information both serve as electronic evidence.

Q.46 Classify Cyber Crimes based on impact. (Unit 1)

Cybercrimes can be classified based on their impact as:

1. **Individual Impact Crimes:** Target personal data, privacy, or identity (e.g., identity theft, cyberstalking).
2. **Corporate Impact Crimes:** Affect organizations through data breaches or ransomware.
3. **National Impact Crimes:** Include cyber-terrorism or attacks on critical infrastructure.

Example: A ransomware attack disrupting hospital systems is a high-impact cybercrime.

Q.47 Give example of Digital Crime Module. (Unit 1)

A digital crime module represents a **specific category or area of cybercrime investigation** focusing on particular offenses or digital environments.

Example: The **Cyber Fraud Investigation Module** focuses on tracing financial crimes such as phishing, online scams, and credit card fraud using forensic and tracking tools.

Q.48 Enlist any two tools for Vulnerability Assessment. (Unit 4)

Two popular tools used for vulnerability assessment are:

1. **Nessus:** Scans systems for known vulnerabilities and misconfigurations.
2. **OpenVAS:** An open-source tool that detects security loopholes in networks and servers.

These tools help organizations identify and fix weaknesses before attackers exploit them.

Example: Nessus detecting outdated software versions on a corporate network.

4 - Mark Questions

Q.1 Distinguish between Hacking and Cracking. (Unit 1)

Basis of Difference	Hacking	Cracking
Definition	Gaining unauthorized access to a computer system or network to explore, test, or exploit vulnerabilities.	Breaking into systems or software to bypass security, remove protections, or cause damage.
Intent	Can be ethical (to test security) or malicious (to steal data).	Always malicious , aimed at violating system integrity or security.
Purpose	To identify weaknesses or enhance security systems.	To steal, modify, or corrupt software and data.
Tools Used	Network scanners, password crackers, exploit frameworks (e.g., Metasploit).	Key generators, serial code breakers, decryption utilities.
Example	Ethical hackers testing a company's firewall for vulnerabilities.	Cracking paid software to use it without a license.

Q.2 Sketch the process of Slack Space Data Retrieval. (Unit 4)

Slack space refers to the **unused storage area** at the end of a file cluster on a hard drive. This area may contain fragments of previously deleted data.

The process of slack space data retrieval involves:

1. **Identification:** Locate the slack space using forensic tools like FTK or EnCase.
2. **Imaging:** Create a forensic image of the drive to prevent altering the original evidence.
3. **Analysis:** Extract and inspect data remnants stored in the slack space.
4. **Recovery:** Retrieve and reconstruct partial files or text fragments from it.
5. **Verification:** Validate the recovered data using hash values to ensure integrity.

Example: Investigators may find fragments of deleted emails in the slack space of a hard drive during forensic analysis.

Q.3 Write short note on Digital Image Processing in Forensics. (Unit 5)

Digital image processing in forensics involves the **analysis and enhancement of images or videos** to extract valuable evidence from digital media. It helps investigators improve the visibility of unclear images, identify objects, detect tampering, and

authenticate visual data.

The process includes **noise reduction, contrast adjustment, filtering, and metadata analysis** to ensure accuracy and authenticity. Investigators can also use image forensics to detect **forged or manipulated content** through pixel analysis and error level examination.

Specialized tools like **Amped FIVE, Adobe Photoshop, and MATLAB** are commonly used for forensic image enhancement.

Example: Enhancing CCTV footage to clearly identify a suspect's face or vehicle number plate during a criminal investigation.

Q.4 Write short note on Vulnerability Assessment Tools. (Unit 4)

Vulnerability assessment tools are used to **identify, evaluate, and report security weaknesses** in networks, systems, or applications. They scan for misconfigurations, outdated software, or exploitable vulnerabilities that could lead to cyberattacks.

These tools provide risk ratings and remediation suggestions to strengthen an organization's defense posture. Common tools include **Nessus, OpenVAS, Nikto, and QualysGuard**.

Example: A vulnerability scan using Nessus detects unpatched software and missing firewall configurations in a corporate network.

Q.5 Examine any two Disk Imaging Tools. (Unit 4)

Disk imaging tools are used to **create exact bit-by-bit copies** of digital storage devices for forensic or backup purposes. Two commonly used tools are:

1. **FTK Imager:** A forensic tool that creates precise images of drives without altering original data. It allows viewing of files and folders and verifies data integrity through hash values.
2. **EnCase:** A commercial forensic suite used for imaging, analyzing, and reporting digital evidence. It supports multiple file systems and offers strong reporting and verification features.

Example: Investigators use FTK Imager to clone a suspect's hard drive before performing forensic analysis.

Q.6 Discuss the role of Cryptography in Cyber Security. (Unit 1)

Cryptography plays a crucial role in **protecting digital data from unauthorized access** and maintaining its confidentiality, integrity, and authenticity. It converts readable data (plaintext) into unreadable form (ciphertext) using encryption algorithms.

In cybersecurity, it is used in secure communication, authentication, and digital signatures. Cryptographic protocols like **SSL/TLS, AES, and RSA** ensure that sensitive information transmitted over networks remains secure.

Example: AES encryption is used to secure online banking transactions and protect customer credentials from interception.

Q.7 Discuss importance of Security Audit in Cyber Laws. (Unit 2)

A **security audit** is essential in cyber laws as it ensures that an organization's information systems comply with legal and regulatory standards for data protection and cybersecurity. It involves assessing network configurations, access controls, and security policies to identify vulnerabilities and prevent cybercrimes.

Through audits, organizations can demonstrate due diligence and accountability in handling sensitive data, thereby reducing the risk of legal penalties. Security audits also help maintain the integrity and confidentiality of information, aligning with the **Information Technology (IT) Act, 2000** and other global standards like **ISO 27001**.

Example: Regular security audits help companies ensure compliance with India's IT Act and protect against data breaches or cyber fraud.

Q.8 Illustrate the concept of Security Assurance. (Unit 2)

Security assurance refers to the **confidence that an organization's systems, processes, and controls are functioning effectively** to protect information assets from threats. It involves continuous monitoring, risk assessments, and compliance checks to verify that security measures meet defined standards.

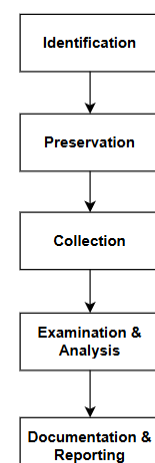
Security assurance ensures **confidentiality, integrity, and availability (CIA)** of information, supporting trust among users and stakeholders. It also helps in identifying weaknesses before attackers can exploit them. Frameworks such as **ISO 27001** and **NIST SP 800-53** guide the assurance process.

Example: Performing periodic penetration tests to verify that a company's firewalls and encryption systems are effective provides strong security assurance.

Q.9 Sketch the architecture of a Forensic Investigation Process. (Unit 5)

The **forensic investigation process** follows a structured architecture designed to ensure accurate and legally valid analysis of digital evidence. The main stages include:

1. **Identification:** Detect potential sources of digital evidence.
2. **Preservation:** Secure and protect evidence using imaging tools to prevent tampering.
3. **Collection:** Acquire data using forensically sound tools like FTK or EnCase.
4. **Examination & Analysis:** Extract relevant information, recover deleted data, and interpret results.



5. **Documentation & Reporting:** Maintain chain-of-custody and prepare detailed reports for legal proceedings.

Example: A forensic expert uses imaging, analysis, and reporting steps to investigate data theft in an organization.

Q.10 Distinguish between FAT and EXT file systems. (Unit 3)

Basis of Difference	FAT File System	EXT File System
Full Form	FAT stands for File Allocation Table , a simple system developed by Microsoft for early DOS and Windows operating systems.	EXT stands for Extended File System , developed for Linux and Unix-based systems to manage large volumes efficiently.
Structure	Uses a table to record clusters of data, which maps files to their physical locations on the disk. Easy to implement but prone to fragmentation.	Uses inodes and block groups to organize data, providing faster access and better recovery capabilities.
Journaling Support	No journaling feature, meaning file corruption can occur after sudden power loss.	EXT3 and EXT4 support journaling , which logs changes before execution, ensuring higher data reliability.
Security Features	Provides minimal file permissions; lacks user-level access control or encryption features.	Supports file permissions, ownerships, and access control lists (ACLs) , ensuring strong security and multi-user management.
Example	Commonly used in pen drives, SD cards, and legacy Windows systems .	Used in Ubuntu, Red Hat, and Android devices for reliable storage and system operations.

Q.11 Examine the functionality of EnCase and FTK tools. (Unit 4)

EnCase and **FTK (Forensic Toolkit)** are two leading digital forensic tools used for data acquisition, analysis, and reporting.

- **EnCase:** Developed by Guidance Software, EnCase provides comprehensive disk imaging, file recovery, and advanced reporting capabilities. It supports multiple file systems (NTFS, FAT, EXT) and ensures data integrity using hash verification. It is widely used in legal and corporate investigations for in-depth forensic analysis.

- **FTK:** Developed by AccessData, FTK offers efficient indexing and keyword search capabilities, helping investigators analyze large datasets quickly. It includes email analysis, registry viewing, and file decryption features.

Example: EnCase can clone a suspect's hard drive, while FTK can rapidly search and filter through thousands of recovered files.

Q.12 Distinguish between FAT and NTFS. (Unit 3)

Basis of Difference	FAT File System	NTFS File System
Full Form	File Allocation Table	New Technology File System
Developer	Developed by Microsoft for early DOS/Windows systems.	Developed by Microsoft for Windows NT and later versions.
Structure	Uses a table to map file locations; simple but inefficient with large disks.	Uses the Master File Table (MFT) to store detailed file and metadata information.
Security	Offers limited security; no user-level access control.	Supports Access Control Lists (ACLs) , file encryption , and permissions for enhanced security.
Reliability	No journaling; data may be lost after improper shutdowns.	Includes journaling and recovery logs , reducing the risk of corruption.
File Size Support	Maximum file size is 4 GB (FAT32) ; partition limit up to 2 TB.	Supports files up to 16 TB and large partition sizes for enterprise use.
Example	Commonly used in USB drives and SD cards .	Used in modern Windows systems and secure internal drives for better performance and security.

Q.13 Examine the use of FTK Tool in Forensic Analysis. (Unit 4)

The **Forensic Toolkit (FTK)**, developed by AccessData, is a comprehensive digital forensics suite used for **data acquisition, analysis, and evidence reporting**. It helps investigators efficiently examine large volumes of data while ensuring the integrity of evidence. Key uses include:

- **Imaging and Data Recovery:** FTK creates bit-by-bit images of storage devices and recovers deleted or hidden files.

- **Indexing and Searching:** Its powerful indexing engine allows rapid keyword searches across emails, documents, and logs.
- **Decryption and Password Recovery:** It can decrypt protected files using integrated password-cracking modules.
- **Evidence Verification:** Uses hashing (MD5/SHA) to maintain chain-of-custody integrity.

Q.14 Discuss different types of File Systems used in Windows. (Unit 3)

Windows operating systems support multiple file systems for data storage and management. The primary types are:

1. **FAT16 (File Allocation Table 16):** Used in older systems; supports smaller partitions up to 2 GB.
2. **FAT32:** Common for USB drives and memory cards; supports up to 4 GB file size.
3. **exFAT:** Designed for flash drives and external storage; supports larger files and better cross-platform compatibility.
4. **NTFS (New Technology File System):** Default for modern Windows systems; supports encryption, compression, journaling, and advanced permissions.

Q.15 Distinguish between FAT16 and FAT32. (Unit 3)

Basis of Difference	FAT16	FAT32
Full Form	File Allocation Table 16-bit	File Allocation Table 32-bit
Partition Size Support	Supports partitions up to 2 GB .	Supports partitions up to 2 TB and beyond.
Cluster Size	Uses larger clusters (typically 32 KB), leading to wasted disk space.	Uses smaller clusters (4–32 KB), allowing more efficient disk usage.
Compatibility	Common in older systems like DOS and early Windows versions.	Supported by modern Windows and portable devices such as USB drives.
File Management Efficiency	Slower and less efficient for large disks.	Improved file allocation efficiency and reduced fragmentation.
Example	Used in MS-DOS systems and early Windows 95 .	Common in USB drives and memory cards for modern systems.

Q.16 Illustrate steps of Security Audit. (Unit 2)

A **security audit** is a structured process that evaluates how well an organization's information systems conform to security standards and policies. The main steps include:

1. **Planning:** Define audit objectives, scope, and standards (e.g., ISO 27001).
2. **Data Collection:** Gather system logs, network configurations, and security policies.
3. **Assessment:** Analyze collected data to identify weaknesses and non-compliance areas.
4. **Testing Controls:** Verify the effectiveness of firewalls, access controls, and encryption mechanisms.
5. **Reporting:** Document findings, highlight vulnerabilities, and suggest corrective actions.
6. **Follow-up:** Ensure remediation steps are implemented successfully.

Example: Conducting a security audit on a company's network to identify unpatched software vulnerabilities.

Q.17 Illustrate handling procedure of Electronic Evidence. (Unit 1)

Handling electronic evidence requires extreme care to ensure it remains authentic, untampered, and admissible in court. The proper procedure involves the following steps:

1. **Identification:** Recognize and label potential digital evidence such as computers, USBs, or mobile devices.
2. **Preservation:** Secure the evidence using anti-static packaging and protect it from alteration.
3. **Collection:** Acquire data using write-blockers and forensic tools (e.g., FTK Imager) to avoid modification.
4. **Documentation:** Record each action taken, including device details and timestamps, to maintain the **chain of custody**.
5. **Analysis and Reporting:** Analyze the acquired data and prepare an evidence report suitable for legal proceedings.

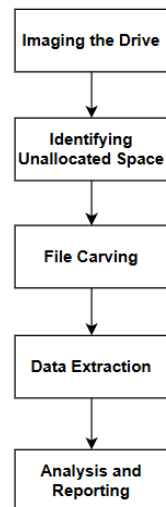
Example: Imaging a suspect's hard drive with FTK Imager while documenting each step for court use.

Q.18 Sketch steps in retrieving deleted data. (Unit 5)

Retrieving deleted data involves recovering files that have been erased but still exist in storage until overwritten. The general steps include:

1. **Imaging the Drive:** Create a bit-by-bit copy of the storage device to preserve original data.
2. **Identifying Unallocated Space:** Locate areas where deleted files may still reside.
3. **File Carving:** Use forensic tools (like Autopsy or EnCase) to reconstruct files based on known headers and footers.
4. **Data Extraction:** Retrieve recovered files and verify integrity using hash values.
5. **Analysis and Reporting:** Examine the restored data for evidential value and include it in forensic reports.

Example: Using Autopsy to recover deleted emails or images from a suspect's laptop during investigation.



Q.19 Examine the role of Forensic Tools in Data Recovery. (Unit 4)

Forensic tools play a crucial role in **retrieving, analyzing, and verifying lost or deleted data** from digital devices while preserving evidence integrity. They help investigators restore files that were accidentally deleted, intentionally erased, or hidden using anti-forensic methods. Key roles include:

1. **Data Acquisition:** Tools like FTK Imager and EnCase create exact disk images for analysis without altering the original media.
2. **File Recovery:** Tools such as Autopsy and R-Studio reconstruct deleted or damaged files.
3. **Data Analysis:** Identify and interpret recovered data relevant to an investigation.
4. **Validation:** Hashing algorithms ensure the recovered data remains authentic.

Example: EnCase is used to recover deleted documents from a formatted hard drive during cyber fraud analysis.

Q.20 Write short note on Data Recovery. (Unit 5)

Data recovery is the **process of restoring lost, deleted, corrupted, or inaccessible information** from digital storage devices such as hard drives, SSDs, memory cards, or mobile phones. It is essential in both digital forensics and disaster recovery.

The process includes **drive imaging, file carving, and metadata reconstruction** to retrieve valuable evidence without altering original data. It helps recover data lost due to accidental deletion, system crashes, or malware attacks.

Forensic experts rely on specialized tools like **Recuva, R-Studio, and Autopsy** to extract and verify recovered information.

Example: Recovering deleted chat messages from a smartphone during a cyberbullying investigation.

Q.21 Illustrate the process of Security Audit. (Unit 2) (Same as Q.16)

Q.22 Write short note on Anti-Forensics. (Unit 4)

Anti-forensics refers to **techniques and tools used to hide, alter, or destroy digital evidence**, making forensic investigations difficult or misleading. The goal is to obstruct evidence recovery, conceal malicious activity, or manipulate digital traces.

Common anti-forensic techniques include **data wiping, encryption, metadata manipulation, steganography, and log tampering**. Investigators counter these methods by using advanced tools capable of detecting anomalies and recovering hidden data.

Example: Using software to securely wipe a hard drive so that deleted files cannot be recovered by forensic tools.

Q.23 Discuss the need of Digital Forensics in Cyber Crime. (Unit 1)

Digital forensics is essential in cybercrime investigations as it enables the **identification, preservation, analysis, and presentation of digital evidence** in a legally acceptable manner. It helps uncover the source, method, and impact of cyber offenses such as hacking, phishing, or data theft. The need of digital forensics:

1. **Recover Deleted or Hidden Data:** Retrieve vital evidence even after attempts to erase it.
 2. **Establish Accountability:** Link digital actions to specific individuals or devices.
 3. **Maintain Evidence Integrity:** Ensure admissibility in court through proper chain-of-custody documentation.
 4. **Prevent Future Attacks:** Provide insights into system vulnerabilities.
-

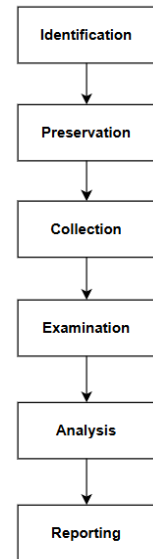
Q.24 Sketch the process of Digital Investigation. (Unit 5)

The **digital investigation process** is a structured sequence of activities designed to uncover and interpret electronic evidence effectively. The key steps include:

1. **Identification:** Recognize potential sources of digital evidence (computers, mobiles, networks).
2. **Preservation:** Secure and protect evidence using imaging and write-blocking methods.

3. **Collection:** Acquire data in a forensically sound manner using tools like FTK or EnCase.
4. **Examination:** Filter and extract relevant data such as deleted files or logs.
5. **Analysis:** Interpret the evidence to reconstruct events and identify culprits.
6. **Reporting:** Document findings clearly and prepare legally admissible reports.

Example: Investigators follow this process to analyze a compromised email server in a cyber fraud case.



7 - Mark Questions

Q.1 Discuss the role of SSE-CMM in Security Assurance. (Unit 2)

The **Systems Security Engineering Capability Maturity Model (SSE-CMM)** is an international framework that defines the best practices and maturity levels for security engineering processes. It ensures that organizations follow consistent, measurable, and repeatable methods for developing and maintaining secure systems, thus strengthening **security assurance**.

SSE-CMM focuses on integrating security into all stages of a system's lifecycle — from design to deployment. It defines **five maturity levels**: *Performed, Managed, Established, Predictable*, and *Optimizing*, which reflect how well an organization's security practices are implemented and improved over time.

By using SSE-CMM, organizations can:

- Evaluate their existing security processes and identify weaknesses.
- Ensure compliance with international security standards like ISO/IEC 27001.
- Improve coordination among developers, security teams, and auditors.
- Establish a foundation for continuous security improvement and risk management.

Example: A government agency can use SSE-CMM to assess the maturity of its cybersecurity policies before deploying national defense information systems.

Q.2 Analyse FAT and NTFS file system differences. (Unit 3)

Aspect	FAT (File Allocation Table)	NTFS (New Technology File System)
Developer & Usage	Developed by Microsoft for early DOS/Windows; still used in portable devices.	Developed by Microsoft for Windows NT and later; standard for modern systems.
Structure	Uses a simple table to track file clusters; prone to fragmentation.	Uses a Master File Table (MFT) storing metadata, enabling faster file access.
Security	Lacks built-in security; no access control or encryption.	Supports ACLs , file encryption, and permission-based security.
Reliability	No journaling; data may be lost after sudden power failure.	Includes journaling and transaction logging , improving data recovery.

File & Volume Support	File size limited to 4 GB ; partition up to 2 TB.	Supports very large files (up to 16 TB) and partitions exceeding 16 TB.
Performance	Lightweight and simple, suitable for USBs or SD cards.	Optimized for large systems, offering compression, encryption, and indexing.
Example	Common in flash drives for compatibility.	Used in enterprise Windows servers for security and stability.

Q.3 Describe the procedure of retrieving deleted data from mobiles. (Unit 5)

Retrieving deleted data from mobile devices is a vital process in **mobile forensics** that helps investigators recover erased information such as messages, contacts, images, and call logs. The procedure must be carried out in a forensically sound manner to ensure evidence integrity and legal admissibility. **Procedure:**

1. **Seizure and Preservation:** The device is secured in a Faraday bag to block network signals and prevent data alteration.
 2. **Imaging:** A bit-by-bit clone of the mobile's storage is created using forensic tools like *Cellebrite UFED* or *XRY*.
 3. **Data Extraction:** Logical or physical extraction methods are used to retrieve both existing and deleted data. Logical focuses on user-accessible data, while physical accesses raw memory, including deleted content.
 4. **File Carving:** Deleted files are reconstructed based on file headers and footers.
 5. **Decryption and Analysis:** Encrypted data is decrypted and examined for evidentiary value.
 6. **Documentation and Reporting:** Findings are documented along with hash values to maintain chain of custody.
-

Q.4 Explain COBIT Framework for IT Governance. (Unit 2)

COBIT (Control Objectives for Information and Related Technologies) is a globally recognized framework for **IT governance and management**, developed by ISACA. It ensures that IT systems align with business goals while maintaining efficiency, compliance, and risk control. **Key Components:**

1. **Principles:** Focus on stakeholder needs, end-to-end governance, and a holistic approach to IT management.
2. **Governance Objectives:** Define processes for planning, building, running, and monitoring IT systems.
3. **Control Processes:** Provide measurable performance metrics for IT resources like applications, data, and infrastructure.

4. **Maturity Model:** Assesses how effectively IT governance is implemented, ranging from ad hoc to optimized levels.
5. **Benefits:** Enhances accountability, risk management, and decision-making transparency.

Example: A financial institution uses COBIT to ensure its IT operations comply with data privacy and cybersecurity regulations.

Q.5 Describe retrieving data from Ghosting and Compressed Files. (Unit 5)

Data retrieval from **ghosting** and **compressed files** is an important process in digital forensics, as valuable evidence may exist within hidden, duplicated, or compacted data.

1. Retrieving Data from Ghosting:

Ghosting refers to the creation of **disk images or cloned copies** of a storage device. Investigators extract data from these ghost images without altering the original evidence.

- The cloned image is analyzed using forensic tools such as *EnCase* or *FTK Imager*.
- Examiners can access deleted, hidden, or residual data sectors.
- Hash verification ensures the cloned image matches the original data to maintain evidentiary integrity.

2. Retrieving Data from Compressed Files:

Compressed files (like .zip, .rar) reduce storage size by encoding multiple files into one container.

- Investigators decompress these files using forensic utilities to access original content.
 - If password-protected, decryption tools like *Passware Kit* are used.
 - File integrity and timestamps are verified to ensure authenticity.
-

Q.6 Examine the process of Data Recovery from Slack Space. (Unit 4)

Slack space is the unused area between the end of a file and the end of its allocated disk cluster. It may contain remnants of previously deleted data, making it valuable for forensic recovery. **Process of Data Recovery:**

1. **Identification:** Locate files and their allocated clusters using forensic tools like *Autopsy* or *EnCase*.
2. **Imaging:** Create a forensic image of the storage device to preserve the original evidence.
3. **Slack Space Analysis:** Examine slack space sectors to detect hidden or partial data fragments.
4. **File Carving:** Recover data by reconstructing files based on known patterns or signatures.

5. **Validation:** Verify recovered data integrity using hash algorithms such as MD5 or SHA-1.
 6. **Reporting:** Document findings and maintain the chain of custody for legal presentation.
-

Q.7 Discuss the challenges in handling Electronic Evidence. (Unit 1)

Handling **electronic evidence** presents multiple challenges due to its volatile nature, ease of manipulation, and complex legal requirements. Maintaining the **authenticity, integrity, and admissibility** of such evidence is critical during investigations.

- 1. Volatility of Data:** Electronic evidence like RAM or cache data is temporary and can be lost if not preserved immediately.
 - 2. Risk of Contamination:** Improper handling or use of non-forensic tools can alter timestamps or file metadata.
 - 3. Large Volume of Data:** Investigators must analyze massive amounts of information, increasing time and resource demands.
 - 4. Encryption and Password Protection:** Encrypted files and devices can delay investigations.
 - 5. Jurisdictional Issues:** Cross-border data storage and international laws complicate evidence access.
 - 6. Maintaining Chain of Custody:** Every step must be documented to prove evidence integrity in court.
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Q.8 Explain International Cyber Law Framework. (Unit 2)

The **International Cyber Law Framework** establishes global norms and regulations to combat cybercrimes and promote secure online practices. It ensures cooperation between nations in addressing cross-border digital offenses. **Key Components:**

1. **Budapest Convention (2001):** The first international treaty on cybercrime, emphasizing investigation cooperation, data sharing, and extradition.
 2. **United Nations (UN) Efforts:** Promotes global cybersecurity awareness and supports frameworks for digital governance.
 3. **European Union's GDPR:** Protects user privacy and regulates cross-border data transfer.
 4. **OECD Guidelines:** Define security and privacy principles for international digital transactions.
 5. **Bilateral and Regional Treaties:** Encourage information sharing and mutual legal assistance among countries.
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Q.9 Explain the importance of Cyber Laws in Digital Security. (Unit 2)

Cyber laws play a vital role in ensuring **digital security, privacy, and accountability** in cyberspace. As the internet becomes integral to daily life, legal frameworks are essential to regulate online behavior and protect individuals and organizations from cybercrimes.

- 1. Protection Against Cybercrimes:** Cyber laws criminalize activities like hacking, identity theft, and online fraud, ensuring offenders are punished under defined statutes.
- 2. Data Protection and Privacy:** Laws such as the **IT Act 2000** (India) and **GDPR** (EU) safeguard personal and financial data against unauthorized use.
- 3. Regulation of E-Commerce:** They provide legal recognition to electronic transactions, contracts, and digital signatures, promoting secure online business.
- 4. Ensuring Cyber Ethics:** Cyber laws promote responsible behavior and discourage misuse of digital resources.
- 5. Legal Admissibility of Evidence:** Defines the procedure for collecting, preserving, and presenting electronic evidence in courts.
- 6. National Security:** Regulates activities that may endanger critical infrastructure or state data.

Example: The Indian IT Act 2000 penalizes unauthorized access to computer systems under Section 66.

Q.10 Illustrate steps in Electronic Evidence Collection. (Unit 1)

Electronic evidence collection involves a systematic and forensically sound process to ensure **authenticity, accuracy, and legal admissibility** of digital data.

- 1. Identification:** Determine potential sources of electronic evidence, such as hard drives, USBs, or cloud storage.
 - 2. Preservation:** Secure devices to prevent tampering or data loss using tools like **write blockers** and Faraday bags.
 - 3. Acquisition:** Create a **bit-by-bit forensic image** of the storage media using tools such as *FTK Imager* or *EnCase*.
 - 4. Verification:** Validate acquired data using cryptographic hash functions (MD5/SHA-1) to ensure integrity.
 - 5. Documentation:** Record all actions, tools used, and timestamps to maintain a clear **chain of custody**.
 - 6. Analysis and Reporting:** Examine data for relevant evidence and prepare comprehensive reports for court submission.
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Q.11 Analyse NTFS structure with its advantages over FAT systems. (Unit 3)

The **NTFS (New Technology File System)** is a robust, secure, and efficient file system developed by Microsoft for modern Windows operating systems. It improves upon FAT by offering enhanced data management, reliability, and access control.

Structure of NTFS:



1. **Boot Sector:** Contains information about the file system structure and boot code.
2. **Master File Table (MFT):** Central database storing details (metadata) about every file, including name, size, permissions, and location.
3. **File Records:** Each file is treated as a record in the MFT, ensuring better organization.
4. **Attributes:** Store file data, security identifiers, timestamps, and compression details.
5. **Journaling:** Logs all changes before execution to prevent data loss during system failures.

Advantages over FAT:

- **Security:** Supports **Access Control Lists (ACLs)** for user-level permissions and **file encryption**.
- **Reliability:** Journaling and transaction logs prevent corruption after crashes.
- **Efficiency:** Handles large volumes and files (up to 16 TB).

Q.12 Examine usage of Slack Space in Disk Storage. (Unit 4)

Slack space refers to the unused storage area between the end of a file and the end of its allocated disk cluster. This space may still contain fragments of previously deleted data, making it highly relevant in digital forensics. **Usage and Importance:**

1. **Data Hiding:** Attackers may intentionally hide information or malicious code within slack space to evade detection.
 2. **Residual Evidence Recovery:** Forensic analysts recover hidden or deleted data remnants using tools like *Autopsy* or *EnCase*.
 3. **Forensic Analysis:** Helps investigators reconstruct partial files or discover hidden evidence in unallocated space.
 4. **Legal Relevance:** Recovered slack data can reveal sensitive information such as fragments of documents or communication logs.
 5. **System Efficiency Insight:** Analysis of slack space helps determine disk usage and data allocation patterns.
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Q.13 Illustrate the steps involved in Digital Forensics Process. (Unit 1)

The **Digital Forensics Process** involves a systematic and legally compliant approach to identify, preserve, analyze, and present digital evidence. It ensures that all procedures maintain the integrity and admissibility of data during investigation. **Steps Involved:**

1. **Identification:** Determine potential sources of evidence such as computers, mobile devices, servers, or cloud storage.
 2. **Preservation:** Secure and protect digital evidence using methods like imaging, write-blocking, and isolation (e.g., Faraday bags).
 3. **Collection:** Acquire data through forensic tools such as *FTK Imager*, *EnCase*, or *Autopsy* to prevent alteration.
 4. **Examination:** Extract relevant files, logs, or deleted information for detailed review.
 5. **Analysis:** Correlate evidence, reconstruct timelines, and identify responsible individuals or entities.
 6. **Documentation:** Record every step of the process, maintaining a **chain of custody** to ensure legal validity.
 7. **Presentation:** Summarize findings in a clear, concise, and court-admissible forensic report.
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Q.14 Examine various Forensic Tools used for Disk Imaging. (Unit 4)

Disk imaging is a critical process in digital forensics that involves creating a **bit-by-bit replica** of digital storage to preserve evidence without modifying the original data.

Various specialized tools are used for this purpose:

1. **FTK Imager:** Creates forensic images in formats like E01 and RAW; allows file previews and hash verification.
2. **EnCase:** Comprehensive suite offering imaging, analysis, and reporting capabilities; ensures evidence authenticity using hash checks.
3. **dd and dc3dd (Linux Tools):** Command-line utilities used to create raw disk images, widely used for open-source forensic work.
4. **Guymager:** Open-source tool that supports multi-threaded imaging and automatic hash generation.
5. **Clonezilla:** Primarily used for large-scale system imaging and duplication in enterprise environments.
6. **X-Ways Forensics:** Lightweight tool supporting multiple file systems, used for quick imaging and data validation.

Example: FTK Imager is used to create a verified image of a suspect's hard drive before performing detailed forensic analysis.

Q.15 Illustrate the process of collecting and storing electronic evidence. (Unit 5)

The **collection and storage of electronic evidence** is a crucial part of digital forensics, as improper handling can compromise its integrity and make it inadmissible in court. The process must follow strict legal and technical guidelines to ensure authenticity and reliability.

- 1. Identification:** Detect potential sources of evidence such as computers, mobile devices, storage media, or network servers.
 - 2. Preservation:** Secure the evidence from alteration or loss using write blockers, anti-static covers, and Faraday bags to prevent network interference.
 - 3. Collection:** Acquire digital data through **bit-by-bit imaging** using forensic tools like *FTK Imager*, *EnCase*, or *Autopsy* without modifying the original media.
 - 4. Verification:** Validate collected evidence using cryptographic hash functions (MD5, SHA-1) to confirm integrity.
 - 5. Documentation:** Record device details, timestamps, methods, and personnel involved to maintain a proper **chain of custody**.
 - 6. Storage:** Preserve evidence in sealed, tamper-proof containers and store it in a secure, access-controlled environment.
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Q.16 Discuss challenges in multimedia forensics. (Unit 5)

Multimedia forensics involves the analysis of digital images, audio, and video files to authenticate content or detect manipulation. However, several challenges make the process complex and resource-intensive.

- 1. Data Volume and Complexity:** High-resolution multimedia files consume large storage and require advanced tools for analysis.
- 2. Tampering and Deepfakes:** Modern editing tools can alter images or videos realistically, making detection difficult.
- 3. Compression Artifacts:** Lossy compression formats (like JPEG or MP4) distort metadata and degrade forensic accuracy.
- 4. Metadata Manipulation:** Attackers can alter timestamps, GPS coordinates, or device details in multimedia files.
- 5. Encryption and Obfuscation:** Encrypted multimedia content requires additional decryption before analysis.
- 6. Tool Limitations:** Forensic tools may struggle to keep pace with emerging multimedia formats and editing technologies.
- 7. Legal and Ethical Barriers:** Admissibility of multimedia evidence depends on authenticity verification standards in court.

Example: Detecting AI-generated deepfake videos during a cyber harassment investigation is a major multimedia forensic challenge.

Q.17 Explain International Security Standards such as ISO/IEC 27001. (Unit 2)

ISO/IEC 27001 is a globally recognized standard for establishing, implementing, maintaining, and improving an **Information Security Management System (ISMS)**. It provides a systematic approach to protecting sensitive organizational information by managing people, processes, and technology. **Key Elements of ISO/IEC 27001:**

1. **Risk Assessment and Treatment:** Identifies information security risks and defines controls to mitigate them.
2. **Security Policy Framework:** Outlines organizational rules and objectives for data protection.
3. **Asset Management:** Ensures that data, hardware, and software are classified and protected appropriately.
4. **Access Control:** Defines user roles and permissions to prevent unauthorized data access.
5. **Incident Management:** Provides guidelines for identifying, reporting, and managing security breaches.
6. **Compliance:** Ensures conformity with legal, regulatory, and contractual requirements.
7. **Continuous Improvement:** Encourages regular reviews and audits for ongoing ISMS enhancement.

Benefits:

- Builds trust with stakeholders and customers.
- Reduces the risk of data breaches and legal non-compliance.
- Enhances organizational resilience and governance.

Example: A financial institution implements ISO/IEC 27001 to secure customer transaction data and meet global compliance standards.

Q.18 Analyse the structure of UNIX File Systems. (Unit 3)

The **UNIX file system (UFS)** is a hierarchical structure that organizes data efficiently for storage, access, and management. It uses the concept of **inodes** and **directories** to store and retrieve files systematically. **Structure Components:**

1. **Boot Block:** Contains essential code to boot the system.
2. **Superblock:** Stores information about the file system, such as size, block count, and status.
3. **Inode Table:** Each file has an inode that holds metadata (permissions, ownership, size, and pointers to data blocks).
4. **Data Blocks:** Store the actual content of files and directories.
5. **Directory Structure:** Organizes files hierarchically, where each directory maps file names to inode numbers.

6. **Journal (in modern UNIX variants):** Records file system changes to prevent corruption after crashes.

Advantages:

- Supports multi-user access and robust permission controls.
- Provides high reliability, security, and fast data access.

Q.19 Analyse the structure and use of NTFS File System. (Unit 3)

The **NTFS (New Technology File System)** is a robust, secure, and efficient file system developed by Microsoft for modern Windows operating systems. It improves upon FAT by offering enhanced data management, reliability, and access control.

Structure of NTFS:



1. **Boot Sector:** Contains information about the file system structure and boot code.
2. **Master File Table (MFT):** Central database storing details (metadata) about every file, including name, size, permissions, and location.
3. **File Records:** Each file is treated as a record in the MFT, ensuring better organization.
4. **Attributes:** Store file data, security identifiers, timestamps, and compression details.
5. **Journaling:** Logs all changes before execution to prevent data loss during system failures.

Uses and Benefits:

- Supports large files and volumes (up to 16 TB).
- Offers encryption, compression, and disk quotas.
- Provides better performance, security, and reliability than FAT systems.

Example: NTFS is widely used in enterprise servers and databases to ensure secure and efficient file storage.

Q.20 Examine Anti-Forensics techniques and counters. (Unit 4)

Anti-Forensics involves deliberate techniques used by cybercriminals to hide, manipulate, or destroy digital evidence, thereby obstructing forensic investigations. Understanding these techniques and their counters is essential for successful forensic analysis. **Common Anti-Forensics Techniques:**

1. **Data Wiping:** Permanently erasing files to make recovery impossible.
2. **Encryption:** Encoding data to restrict access to authorized users only.

3. **Steganography:** Hiding data within images, audio, or video files.
4. **Metadata Manipulation:** Changing timestamps, authorship, or file creation details.
5. **Log Tampering:** Deleting or altering system logs to cover tracks.
6. **Use of Anonymous Networks:** Employing VPNs or Tor to hide digital footprints.

Counters to Anti-Forensics:

- **Use of Advanced Tools:** Tools like *Autopsy*, *EnCase*, and *X-Ways* can detect and recover hidden or encrypted data.
- **Hash Verification:** Ensures evidence integrity.
- **Regular Training:** Forensic experts must stay updated on emerging evasion techniques.

Example: A forensic analyst uses steganalysis tools to uncover hidden text embedded within an image file during an investigation.

Q.21 Describe how deleted data is recovered from Laptops. (Unit 5)

Recovering deleted data from laptops is an essential part of **digital forensics** that helps investigators retrieve evidence even after files have been erased. When data is deleted, only its file reference in the file system is removed, while the actual content remains in unallocated space until overwritten. **Steps in Recovery:**

1. **Preservation:** Secure the laptop to prevent further data alteration. A forensic image of the hard drive is created using tools like *FTK Imager* or *EnCase*.
2. **Analysis of File System:** Examine file system structures such as FAT, NTFS, or EXT to locate deleted file entries and unallocated clusters.
3. **File Carving:** Reconstruct deleted files based on known headers and footers using tools like *Autopsy* or *R-Studio*.
4. **Metadata Examination:** Recover file names, creation dates, and user access details from file system metadata.
5. **Validation:** Verify recovered files using hash functions (MD5/SHA-1) to confirm authenticity.
6. **Reporting:** Document all recovery procedures and findings for legal submission.

Q.22 Describe recovery of data from compressed files. (Unit 5)

Recovering data from **compressed files** is vital in digital forensics because suspects often compress files to conceal or transfer data efficiently. Compression reduces file size and may involve encryption or password protection.

Procedure for Recovery:

1. **Identification:** Locate compressed files (e.g., .zip, .rar, .7z) during forensic imaging or analysis.
2. **Extraction:** Use tools like *WinRAR*, *7-Zip*, or *FTK* to extract contents safely without modifying original data.

3. **Password Cracking (if encrypted):** Employ password recovery tools such as *Passware Kit* or *John the Ripper* to unlock protected archives.
4. **File Carving:** If files are partially corrupted, carving techniques are used to reconstruct internal data segments.
5. **Integrity Verification:** Use checksums or hash values to ensure extracted files match the originals.
6. **Analysis:** Review extracted data for hidden or suspicious content, such as steganographic files or logs.
7. **Documentation:** Record the extraction process to maintain chain of custody.

Q.23 Discuss the relevance of Web Technology in Cyber Crimes. (Unit 1)

Web technology plays a crucial role in both enabling and combating **cybercrimes**. The internet's open architecture and global connectivity have created opportunities for communication, commerce, and data sharing—but also for exploitation by cybercriminals.

1. **Platform for Cybercrimes:** Websites, social media, and web applications are common platforms for phishing, data theft, and identity fraud. Attackers exploit vulnerabilities in web servers and browsers to gain unauthorized access.
2. **Malware Distribution:** Malicious scripts and drive-by downloads on infected websites are used to install malware on users' systems.
3. **Phishing and Social Engineering:** Fake web portals mimic legitimate sites to steal login credentials or financial information.
4. **Financial Crimes:** E-commerce platforms and online payment gateways can be exploited for credit card fraud and money laundering.
5. **Data Breaches:** Poorly secured web applications may leak sensitive information.
6. **Cyber Investigation Role:** Conversely, forensic experts use web technologies to trace IP addresses, monitor logs, and collect electronic evidence.

Example: Phishing websites posing as banking portals are used to steal customers' credentials, later traced using digital forensic analysis.

Q.24 Illustrate steps in handling and preservation of Electronic Evidence. (Unit 1)

The handling and preservation of **electronic evidence** are critical to ensure that data remains authentic, untampered, and admissible in court. Digital evidence is fragile and can easily be modified or destroyed if not properly managed. **Steps Involved:**

1. **Identification:** Locate potential evidence such as hard drives, mobile devices, or storage media at the crime scene.
2. **Preservation:** Secure evidence immediately using Faraday bags or power disconnection to prevent remote tampering.
3. **Documentation:** Record device details, timestamps, serial numbers, and photographs to maintain authenticity.

4. **Collection:** Acquire forensic images of the data using tools like *FTK Imager* or *EnCase* to prevent modification of the original media.
 5. **Chain of Custody:** Maintain a detailed log of every person who handles the evidence.
 6. **Storage:** Store evidence in tamper-proof, access-controlled environments to ensure safety.
 7. **Verification:** Use hash values to confirm data integrity throughout the investigation process.
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12 - Mark Questions

Q.1 Describe steps of complete Digital Investigation Cycle. (Unit 5)

The **Digital Investigation Cycle** is a systematic and structured process used to identify, collect, analyze, and present digital evidence in a legally admissible manner. It ensures that every step maintains the **authenticity, integrity, and reliability** of digital evidence. The process is cyclical, emphasizing continuous improvement and documentation throughout the investigation.

1. Identification

The first step involves **recognizing potential sources of digital evidence**. Investigators identify devices or locations where evidence may reside, such as computers, mobile phones, cloud storage, USB drives, or network logs.

- It also includes understanding the nature of the incident—data theft, hacking, or fraud—and the scope of investigation.
- Proper planning and resource allocation are done in this stage.

2. Preservation

Preservation ensures the **protection of evidence** from alteration, damage, or loss.

- Investigators isolate affected systems using **Faraday bags** (for wireless devices) or by disconnecting from the network to prevent tampering.
- Write blockers are used to prevent modification while accessing data.
- Every action taken is logged to maintain the **chain of custody**.

3. Collection

This phase focuses on **acquiring digital evidence** in a forensically sound manner.

- A **bit-by-bit image** of storage media is created using tools like *FTK Imager*, *EnCase*, or *Autopsy*.
- Data from volatile memory (RAM) and external devices is also collected.
- Integrity is verified using cryptographic hash functions (MD5, SHA-1).

4. Examination

The collected data is **filtered, organized, and prepared for analysis**.

- Temporary, deleted, or hidden files are recovered.
- File systems (FAT, NTFS, EXT) and metadata are examined.
- Investigators look for relevant artifacts such as browser history, email logs, and registry files.

5. Analysis

This step involves **interpreting the evidence** to establish timelines, user actions, or connections to suspects.

- Correlation between different data sources is established.

- Tools like *X-Ways*, *Cellebrite*, or *Autopsy* assist in pattern recognition and report generation.
- Findings must support hypotheses based on the case objective.

6. Documentation and Reporting

Every step in the investigation is **documented thoroughly** for legal validation.

- The report includes evidence sources, tools used, data recovered, and conclusions.
- The **chain of custody** is attached to prove authenticity.
- Reports are clear, concise, and suitable for non-technical legal audiences.

7. Presentation

In this stage, the evidence and findings are **presented in court or to authorities**.

- Expert witnesses explain technical aspects in simple terms.
- Demonstrations, visual aids, and metadata logs support arguments.
- Investigators must defend procedures to prove data integrity.

Q.2 Describe Data Recovery from Damaged SIM. (Unit 5)

A **Subscriber Identity Module (SIM)** card is a small, removable smart card that stores user identity, network information, contact lists, SMS messages, and authentication keys. In digital forensics, SIM cards are valuable sources of evidence in cases like cyber fraud, identity theft, or mobile-based crimes. However, when a SIM is physically or logically damaged, specialized **data recovery techniques** are required to retrieve vital information without altering the original data. **Data Recovery Process:**

Step 1: Initial Assessment and Preservation

- Inspect the SIM card for physical or electronic damage.
- Clean the surface gently to remove debris or corrosion.
- Document the condition and store it in an anti-static container to prevent further damage.

Step 2: Imaging the SIM

- Create a **forensic image** of the SIM to avoid handling the original directly.
- Tools such as *SIMCon*, *MOBILedit Forensic*, or *Cellebrite UFED* are used to extract binary data from readable memory sectors.
- Hash values (MD5/SHA-1) are generated to ensure the authenticity of the image.

Step 3: Physical Data Extraction

If the SIM is **physically damaged**, chip-off or micro-probing techniques are employed:

- **Chip-Off Technique:** The memory chip is detached and read using specialized hardware.
- **Micro-Probing:** Tiny probes are used to read data directly from the SIM chip's circuits without complete removal.

- Extracted data is stored in a raw format for later decoding.

Step 4: Logical Data Recovery

If the SIM is readable but corrupted:

- Logical extraction tools access structured files like EF_SMS or EF_ADN (contacts).
- Data is decoded according to GSM/3GPP standards.
- Damaged file sectors are reconstructed using software algorithms that map binary patterns to known file structures.

Step 5: Data Analysis

Recovered data may include:

- **Subscriber Information:** IMSI, ICCID, authentication keys.
- **User Data:** SMS, call logs, and contact lists.
- **Network Details:** Last connected tower and timestamp information.

The extracted data is cross-verified with mobile device logs or network records to strengthen the investigation.

Step 6: Documentation and Reporting

All steps, tools used, and findings are thoroughly documented. The **chain of custody** is maintained, and a formal forensic report is generated for court submission.

Q.3 Describe the process of retrieving multimedia evidence. (Unit 5)

Multimedia evidence refers to **digital audio, video, and image files** that serve as crucial proof in forensic investigations. Such evidence is often extracted from computers, mobile devices, CCTV systems, social media, or cloud platforms. The process of retrieving multimedia evidence involves **identifying, preserving, extracting, authenticating, and analyzing** digital content while ensuring its integrity and admissibility in court.

1. Identification of Multimedia Sources

The first step is to identify potential **sources of multimedia evidence**, which may include:

- **Digital Devices:** Smartphones, laptops, DVRs, cameras, and external drives.
- **Online Platforms:** Social media, emails, and cloud storage.
- **IoT Devices:** Smart home security systems or dashcams.

Investigators determine the relevance of each source to the case and create an evidence acquisition plan.

2. Preservation of Multimedia Evidence

Preserving evidence ensures that files remain unaltered from their original form.

- Devices are disconnected from networks to prevent tampering.

- Forensic imaging tools (e.g., *FTK Imager*, *EnCase*) create exact replicas of drives or memory cards.
 - Hash values (MD5/SHA-1) are generated to validate integrity.
- Proper preservation prevents loss due to overwriting, compression, or editing.

3. Extraction of Multimedia Data

Multimedia data is extracted using forensic tools and methods tailored to file types and devices.

- **Logical Extraction:** Retrieves accessible data such as stored videos, photos, or audio recordings.
- **Physical Extraction:** Acquires all data, including deleted or hidden files, by reading memory sectors.
- **Metadata Extraction:** Gathers details like timestamps, GPS data, and camera models from EXIF metadata in images.

4. Authentication of Multimedia Evidence

Authenticating evidence ensures that it hasn't been tampered with or altered.

- **Hash Verification:** Confirms file integrity by comparing hash values before and after extraction.
- **Metadata Analysis:** Checks timestamps, geolocation, and device information for inconsistencies.
- **Error Level Analysis (ELA):** Detects areas in images that may have been digitally manipulated.
- **Frame Analysis:** Identifies frame drops or editing traces in video evidence.

5. Analysis and Enhancement

Once retrieved and verified, multimedia files undergo detailed forensic analysis:

- **Audio Forensics:** Noise reduction, voice identification, and spectrogram analysis.
- **Video Forensics:** Frame-by-frame analysis, facial recognition, and motion tracking.
- **Image Forensics:** Detection of digital alterations, steganography, and pixel-level analysis.

Tools like *Amped FIVE*, *Adobe Photoshop Forensics Toolkit*, and *VLC Metadata Analyzer* assist in evidence enhancement and validation.

6. Documentation and Reporting

All procedures, tools, and findings are carefully documented.

- The **chain of custody** is maintained to record each handler and process.
- Reports include extracted metadata, file integrity details, and analytical conclusions.
- Visual exhibits (e.g., screenshots or frame captures) support the written findings.

7. Presentation of Multimedia Evidence

In court, multimedia evidence must be presented in a **clear, authentic, and comprehensible format**.

- Investigators demonstrate how the evidence was retrieved and authenticated.
 - Technical jargon is simplified for judges and juries.
 - Any enhancements are explained to ensure transparency.
-

Q.4 Describe the process of retrieving multimedia and SIM data in forensics. (Unit 5)

In digital forensics, the retrieval of **multimedia and SIM data** plays a crucial role in uncovering vital evidence related to criminal, corporate, or cyber incidents. Multimedia evidence includes **images, audio, and video files**, while SIM data provides **subscriber identity, call logs, SMS, and network information**. Both types of data are often critical in linking suspects to criminal activities or digital communications. The retrieval process requires specialized tools and techniques to ensure data integrity, accuracy, and admissibility in court.

1. Retrieval of Multimedia Data

Multimedia data can originate from various digital devices such as smartphones, computers, cameras, and CCTV systems. Its recovery must be performed using forensically sound methods.

Step 1: Identification and Preservation

- Identify devices containing multimedia data — including hard drives, DVRs, or mobile storage.
- Preserve data integrity by isolating devices and creating **forensic images** using tools like *FTK Imager* or *EnCase*.
- Generate **hash values** (MD5, SHA-1) to verify authenticity.

Step 2: Data Extraction

- Use forensic software such as *Cellebrite UFED*, *Autopsy*, or *Amped FIVE* to extract image, video, and audio files.
- Apply **logical extraction** for accessible data and **physical extraction** for deleted or hidden files.
- Retrieve **metadata** (e.g., timestamps, GPS data, and device details) for contextual information.

Step 3: Analysis and Authentication

- Verify file authenticity using hash verification and metadata comparison.
- Conduct **image and video forensics** to detect tampering (e.g., Error Level Analysis or frame integrity check).
- Enhance multimedia content (e.g., noise reduction, brightness correction) for clearer identification.

Step 4: Documentation

- Record the extraction methods, software used, and all findings.
 - Maintain a detailed **chain of custody** for court presentation.
-

2. Retrieval of SIM Data

SIM cards store **subscriber and communication data** essential in mobile forensics. Retrieving data from damaged or intact SIM cards provides information on user identity and activity.

Step 1: Examination and Imaging

- Inspect the SIM card for physical or electronic damage.
- Create a **forensic image** using tools like *MOBILedit Forensic*, *SIMCon*, or *Cellebrite UFED*.
- Generate and document hash values to ensure evidence integrity.

Step 2: Logical and Physical Data Extraction

- **Logical Extraction:** Recovers accessible data such as contacts, SMS, and IMSI numbers.
- **Physical Extraction:** Retrieves deleted or hidden data directly from the memory chip using chip-off or micro-probing methods.
- Extracted data may include **ICCID, IMSI, SMS messages, call logs, and last tower location**.

Step 3: Data Decoding and Reconstruction

- Decoded data is analyzed using GSM/3GPP standards.
- Reconstructed data can link suspects to specific devices, locations, or communication events.
- Cross-verification is done with network service provider logs.

Step 4: Reporting and Preservation

- All recovered data is documented with timestamps, extraction tools, and verification records.
 - The SIM card and recovered data are stored in tamper-proof evidence bags.
-

Q.5 Explain process of Disk Imaging and Data Recovery. (Unit 4)

In digital forensics, **Disk Imaging and Data Recovery** are two of the most crucial processes used to preserve and analyze digital evidence. Disk imaging involves creating an **exact, bit-by-bit replica** of a storage device without altering its original data, while data recovery focuses on retrieving **deleted, corrupted, or hidden files** from the storage media. These processes ensure that investigations are conducted in a forensically sound manner while maintaining the integrity and admissibility of the evidence.

1. Disk Imaging Process

Disk imaging is performed to duplicate a digital storage medium, such as a hard drive, SSD, or USB, ensuring the original remains unaltered during examination.

Step 1: Preparation

- **Device Isolation:** Disconnect the suspect device from networks to prevent remote interference.
- **Write Blockers:** Use hardware or software write blockers to prevent any modification to the source disk.
- **Documentation:** Record details like serial numbers, device model, and initial hash values (MD5/SHA-1) for integrity verification.

Step 2: Creating the Forensic Image

- **Bit-by-Bit Copy:** A sector-by-sector copy of the disk is made, including deleted and hidden sectors.
- **Tools Used:** Common imaging tools include *FTK Imager*, *EnCase*, *Guymager*, and *dd/dc3dd*.
- **Imaging Formats:** The image can be saved in formats like RAW (.dd), E01 (EnCase), or AFF.

Step 3: Verification

- The integrity of the image is verified using **hashing algorithms** (MD5/SHA-256).
- Matching hash values between the source and image confirm a successful, tamper-proof duplication.

Step 4: Secure Storage

- The original disk is stored in a **tamper-proof evidence locker**.
 - Analysis is conducted only on the cloned image to preserve the original evidence.
-

2. Data Recovery Process

Data recovery aims to retrieve files that have been **deleted, corrupted, or hidden** from digital storage. Forensic recovery focuses on maintaining legal admissibility and data authenticity.

Step 1: Identification of Data Loss

Determine the cause of data loss — accidental deletion, formatting, corruption, malware, or hardware failure.

Step 2: Logical Data Recovery

- Involves recovering data from an intact file system (FAT, NTFS, EXT, etc.).
- Tools like *Recuva*, *R-Studio*, and *Autopsy* can reconstruct deleted files based on metadata and file headers.

- **File Carving:** Files are reconstructed by identifying patterns (headers and footers) even when directory entries are missing.

Step 3: Physical Data Recovery

- Applied when the device is **physically damaged** or has firmware issues.
- Specialists use cleanroom environments to repair drives and extract raw data directly from platters or chips.
- Specialized tools like *PC-3000* or *DeepSpar Disk Imager* are used for hardware-level recovery.

Step 4: Verification and Validation

Recovered data is compared with hash values or file integrity checks to ensure accuracy and authenticity.

Step 5: Analysis and Reporting

Recovered files are analyzed for case relevance. Findings are documented with timestamps, extraction tools, and validation methods. Reports must be clear enough for **court presentation**.

Q.6 Explain Digital Investigation Process. (Unit 5) (Same as Q.1)

Q.7 Explain Data Acquisition Techniques used in Forensics. (Unit 3)

Data acquisition in digital forensics refers to the process of **collecting digital evidence** from various electronic sources while maintaining its integrity and admissibility. The goal is to obtain a **bit-by-bit replica** or verified copy of the original data without altering it. This step forms the foundation of forensic analysis, as improperly acquired data can render evidence invalid in court.

1. Objectives of Data Acquisition

- Preserve the original data without modification.
- Create a verified duplicate using cryptographic hashes (MD5/SHA-1).
- Ensure forensic soundness through proper documentation.
- Maintain **chain of custody** to support legal admissibility.

2. Types of Data Acquisition Techniques

a) Static (Offline) Acquisition

- Performed when the system is **powered off**.
- Investigators remove storage media (hard drives, SSDs, or USBs) and use **write blockers** to prevent changes.
- A **bit-stream image** is created using tools such as *FTK Imager* or *Guymager*.
- This method provides a complete, unaltered copy of the drive, including deleted and hidden files.

Advantages:

- Ensures data integrity and minimizes the risk of alteration.
- Ideal for detailed examination in controlled environments.

Example: Imaging a suspect's hard drive from a seized desktop computer before analysis.

b) Live (Online) Acquisition

- Conducted while the system is **powered on** to capture volatile data such as RAM, running processes, network connections, and encryption keys.
- Volatile data disappears once the device is shut down, so this method is crucial in incident response and real-time investigations.
- Tools like *Belkasoft Live RAM Capturer* or *Volatility Framework* are used.

Advantages:

- Captures in-memory data, including encryption keys or open sessions.
- Essential for investigations involving live servers or cloud environments.

Example: Capturing the RAM of a running server to extract active user sessions in a cyberattack case.

c) Logical Acquisition

- Extracts **user-level data** such as files, documents, and logs without accessing unallocated or deleted sectors.
- Commonly used for smartphones and operating systems where full imaging is restricted.
- Tools such as *Cellebrite UFED* and *XRY* perform logical extractions from mobile devices.

Advantages:

- Faster than bit-by-bit imaging.
- Useful for cases where full access isn't possible (e.g., encrypted systems).

Example: Extracting call logs and messages from a mobile phone without altering its data.

d) Physical Acquisition

- Captures **the entire storage medium**, including deleted, hidden, or unallocated sectors.
- Produces a raw image that can be examined for hidden partitions or slack space.
- Often used when the forensic examiner requires **deep-level analysis**.

Advantages:

- Provides a complete and exact copy of the device.
- Enables recovery of deleted or partially overwritten data.

Example: Imaging an entire SSD to recover deleted financial records.

e) Targeted or Selective Acquisition

- Used when only specific files or folders are relevant to the investigation.
- Ideal for large storage volumes or time-sensitive investigations.
- Forensic tools filter data based on keywords, file types, or timestamps.

Advantages:

- Saves time and reduces data overload.
- Useful for corporate investigations or legal discovery cases.

Example: Extracting only emails and Excel files from a company's network server for audit purposes.

3. Verification and Validation

After acquisition, integrity verification ensures that no data was altered during the process.

- **Hash Verification:** The hash value (MD5/SHA-256) of the original and copied data must match.
 - **Documentation:** Each step, tool, and personnel involved are logged to maintain chain of custody.
-

Q.8 Explain in detail the Data Acquisition and Authentication Process. (Unit 3)

In **digital forensics**, the **Data Acquisition and Authentication process** is a foundational procedure that ensures the secure collection and validation of digital evidence. Data acquisition refers to the process of **capturing and duplicating electronic data** from various storage media or devices, while **authentication** verifies that the collected data remains **unaltered and genuine**. Both processes are critical for maintaining the **integrity, reliability, and admissibility** of evidence in legal proceedings.

1. Objectives of Data Acquisition and Authentication

- Preserve original data without alteration.
- Create accurate, verifiable copies of evidence.
- Ensure forensic soundness through controlled methods.
- Validate evidence integrity using cryptographic hashing.
- Maintain a clear and continuous **chain of custody**.

2. Data Acquisition Process

The data acquisition process involves systematically collecting data from digital sources using forensic tools and hardware under strict procedural control.

Step 1: Preparation

- Identify the type of device (hard drive, mobile phone, SSD, cloud storage, etc.).

- Use appropriate **forensic hardware and software** tools such as *FTK Imager*, *EnCase*, *Cellebrite UFED*, or *Guymager*.
- Document the device details, including serial numbers and timestamps, before acquisition.

Step 2: Isolation and Preservation

- The device is disconnected from networks to prevent tampering or remote access.
- A **write blocker** is used to prevent any modification during data copying.
- Forensic imaging is performed rather than direct analysis on the original evidence.

Step 3: Creating a Forensic Image

- A **bit-by-bit image** of the storage medium is created to ensure all data, including deleted or hidden sectors, is captured.
- Common formats: RAW (.dd), E01 (EnCase), AFF.
- Tools like *dd*, *FTK Imager*, and *Guymager* generate images with accompanying metadata and hash values.

Step 4: Verification

- The **hash values (MD5, SHA-256)** of the original and the image are compared.
- If both hashes match, it confirms a **forensically sound copy**.
- The hash serves as a digital fingerprint of the evidence.

Step 5: Storage and Documentation

- The original evidence is sealed and stored in a **tamper-proof evidence locker**.
- The forensic image is used for examination and analysis.
- Detailed logs, including tool versions and operator information, are recorded.

3. Authentication Process

Authentication ensures that the collected digital evidence is **genuine, unmodified, and legally valid**. It confirms the **integrity and authenticity** of data throughout the forensic process.

Step 1: Hash Value Generation

- Cryptographic algorithms like **MD5, SHA-1, or SHA-256** generate unique hash values for each piece of evidence.
- Even a minor data change results in a different hash, ensuring tamper detection.

Step 2: Chain of Custody Maintenance

- The chain of custody records each handler, location, and transfer of evidence.
- Any gap can question authenticity in legal proceedings.

Step 3: Verification during Analysis

- Every time evidence is accessed, the current hash value is compared with the original to confirm integrity.
- Tools automatically log hash comparisons to maintain authenticity logs.

Step 4: Reporting

- Hash results and integrity verification steps are included in the final forensic report.
- Documentation ensures transparency and compliance with standards like **ISO/IEC 27037** (Guidelines for digital evidence handling).

Example: The original and image hashes remain consistent before and after analysis, confirming the data's integrity for court submission.

Q.9 Rewrite about the following: (Unit 2)

a) COBIT

b) SSE-CMM

c) Security Audit

a) COBIT (Control Objectives for Information and Related Technologies)

COBIT is a globally recognized **IT governance and management framework** developed by ISACA (Information Systems Audit and Control Association). It provides a comprehensive model that helps organizations ensure their **information systems are efficient, secure, and aligned with business goals**.

Key Features of COBIT:

1. **Framework for Governance:** Ensures that IT operations support and enable business objectives while maintaining compliance and risk management.
2. **Process Model:** Divides IT management into domains — *Plan and Organize, Acquire and Implement, Deliver and Support, Monitor and Evaluate*.
3. **Control Objectives:** Defines specific measurable goals for IT governance and performance evaluation.
4. **Performance Measurement:** Uses maturity models and metrics to assess process capability and effectiveness.
5. **Integration with Standards:** Works in harmony with ISO 27001, ITIL, and other international standards.

Benefits:

- Enhances decision-making transparency.
- Improves accountability for IT processes.
- Reduces operational and security risks.

Example: A bank uses COBIT to monitor IT compliance with financial data regulations and ensure secure transaction systems.

b) SSE-CMM (Systems Security Engineering – Capability Maturity Model)

SSE-CMM is an international model developed to evaluate and improve the **maturity of an organization's security engineering processes**. It establishes standardized practices for implementing and managing information security throughout a system's lifecycle.

Structure of SSE-CMM:

1. **Process Areas:** Defines core security engineering processes such as risk assessment, system design, and incident management.
2. **Maturity Levels:** Comprises five levels – *Performed, Managed, Established, Predictable, and Optimizing*, each indicating process improvement and consistency.
3. **Process Evaluation:** Measures the organization's ability to engineer secure systems and continuously improve its security practices.

Objectives:

- Integrate security into system development from the design stage.
- Ensure consistent and measurable security practices.
- Align with international security standards like ISO/IEC 27001.

Benefits:

- Enhances the security posture of organizations.
- Promotes continuous process improvement and risk mitigation.

Example: A defense organization applies SSE-CMM to assess its software development lifecycle security maturity before deploying mission-critical systems.

c) Security Audit

A **Security Audit** is a systematic evaluation of an organization's **information systems, policies, and practices** to determine how well they comply with established security standards and protect against potential risks.

Process of Security Audit:

1. **Planning:** Define audit objectives, scope, and resources.
2. **Information Gathering:** Collect system configurations, access logs, and security policy documents.
3. **Assessment:** Evaluate system vulnerabilities, user privileges, and adherence to security protocols.
4. **Testing:** Conduct vulnerability scans and penetration tests using tools like *Nessus* or *OpenVAS*.

5. **Reporting:** Document findings, highlight risks, and provide mitigation recommendations.

Types of Security Audits:

- **Internal Audit:** Conducted by in-house teams to assess internal controls.
- **External Audit:** Performed by third-party auditors to ensure compliance with standards like ISO 27001, GDPR, or HIPAA.

Importance:

- Identifies and mitigates security weaknesses.
- Ensures legal and regulatory compliance.
- Builds trust among stakeholders by demonstrating data protection.

Example: A healthcare firm undergoes a security audit to ensure compliance with HIPAA regulations and safeguard patient data.

Q.10 Rewrite about the following: (Unit 3)

a) Electronic Media

b) OS Artifacts

c) Internet Artifacts

a) Electronic Media

Electronic media refers to any device or medium that stores or transmits digital data in electronic form. It includes physical storage devices and digital communication platforms that are critical sources of **digital evidence** in forensic investigations. Investigators use these media to extract information such as files, logs, or communications that can establish a chain of events or identify criminal activities.

Types of Electronic Media:

1. **Primary Storage Devices:** Hard drives (HDDs), Solid-State Drives (SSDs), and removable flash drives that contain operating systems, user data, and system logs.
2. **Secondary Storage Devices:** CDs, DVDs, and external drives used for data backup or archival storage.
3. **Communication Media:** Emails, instant messaging platforms, and social media used for transmitting data.
4. **Cloud and Network Storage:** Online data repositories such as Google Drive, OneDrive, and network-attached storage (NAS) systems.
5. **Mobile and IoT Devices:** Smartphones, SIM cards, and smart gadgets containing location, call, and app data.

Importance in Forensics:

- Serves as a repository of potential evidence such as deleted files, hidden data, and timestamps.
- Helps establish user behavior and event timelines.
- Provides crucial metadata (e.g., last access time, file owner).

Example: Forensic experts recover financial transaction logs from a suspect's USB drive to prove data theft.

b) OS Artifacts (Operating System Artifacts)

OS Artifacts are data remnants created by an operating system during its routine operation. These artifacts provide valuable insights into user activities, system configurations, and historical events on a digital device. Forensic investigators analyze these artifacts to reconstruct user behavior and system interactions.

Common OS Artifacts:

1. **System Logs:** Record system events such as login times, application usage, and system errors.
2. **Registry Files (Windows):** Store configuration data, recently accessed files, and connected USB devices.
3. **Prefetch Files:** Contain information about executed programs, helping to identify frequently used applications.
4. **Recent and Temporary Files:** Help determine which documents or files were opened recently.
5. **Recycle Bin Data:** Stores deleted file references before permanent removal.
6. **Event Viewer Logs:** Record system, application, and security events for auditing.

Forensic Importance:

- Tracks user activities and login patterns.
- Helps identify malware installations or unauthorized access.
- Supports timeline reconstruction during investigations.

Example: Investigators use Windows Registry and Prefetch files to prove that a hacking tool was executed on a specific date.

c) Internet Artifacts

Internet artifacts are digital traces left behind by users while browsing or interacting online. They include cached files, cookies, download histories, and browsing records that reveal user habits, communication, and web-based activities. These artifacts are vital in **cybercrime investigations**, such as phishing, identity theft, and online fraud.

Common Types of Internet Artifacts:

1. **Browser History:** Lists URLs visited, timestamps, and session durations.

2. **Cookies:** Small text files storing user preferences, login data, and session identifiers.
3. **Cache Files:** Store temporary copies of web pages, images, and scripts for faster access.
4. **Download Logs:** Record filenames, sources, and timestamps of downloaded content.
5. **Search Engine Queries:** Provide insight into user interests or criminal intent.
6. **Email and Chat Logs:** Contain communications that may include incriminating evidence.

Forensic Importance:

- Reveals user intent and behavioral patterns.
- Helps trace communication between suspects.
- Identifies malicious websites or illegal downloads.
- Can correlate online activity with system logs for timeline analysis.

Tools Used:

- *Browser History Examiner, Autopsy, Internet Evidence Finder (IEF), and FTK.*

Example: Investigators retrieve browser cache and cookies showing the suspect's access to a phishing website used in financial fraud.

Q.11 Rewrite about the following: (Unit 2)

a) IPR (Intellectual Property Rights)

b) COBIT (Control Objectives for Information and Related Technologies)

c) Security Laws

a) IPR (Intellectual Property Rights)

Intellectual Property Rights (IPR) are the legal rights granted to individuals or organizations for creations of the mind—such as inventions, literary works, symbols, designs, and software. In the context of computer forensics and cybersecurity, IPR plays a key role in **protecting digital assets, software, and data ownership**, thereby promoting innovation while preventing unauthorized use or duplication.

Types of Intellectual Property Rights:

1. **Copyright:** Protects original creative works like software, code, documentation, and multimedia content.
2. **Patents:** Provide exclusive rights for inventors over new technologies, algorithms, or hardware innovations.
3. **Trademarks:** Protect brand identity elements such as names, symbols, and logos used in commerce.
4. **Trade Secrets:** Safeguard confidential business information like formulas, source code, and customer databases.

5. **Industrial Design Rights:** Protect aesthetic designs and user interface structures.

Importance in Cybersecurity:

- Prevents digital piracy and software plagiarism.
- Protects developers and organizations from intellectual theft.
- Encourages ethical innovation and fair competition.
- Supports forensic experts in determining data or code ownership during legal disputes.

Example: A company files an IPR case after detecting its proprietary source code copied and redistributed online without authorization.

b) COBIT (Control Objectives for Information and Related Technologies)

COBIT, developed by **ISACA (Information Systems Audit and Control Association)**, is a globally recognized **IT governance and management framework**. It helps organizations align IT processes with business goals, ensuring that information systems are secure, reliable, and compliant with regulations.

Key Components of COBIT:

1. **Framework Principles:** Focuses on stakeholder needs, end-to-end governance, and comprehensive coverage of IT processes.
2. **Domains and Processes:** Includes domains like *Plan and Organize*, *Acquire and Implement*, *Deliver and Support*, and *Monitor and Evaluate*.
3. **Performance Measurement:** Uses maturity models and key performance indicators (KPIs) to evaluate IT efficiency and risk management.
4. **Integration:** Works in conjunction with other standards such as ISO/IEC 27001 and ITIL.

Benefits:

- Enhances data security and process control.
- Improves accountability and transparency in IT operations.
- Assists organizations in complying with regulatory frameworks.
- Promotes better risk assessment and governance practices.

Example: A financial institution applies COBIT to ensure IT compliance with data protection standards like GDPR and to improve internal security controls.

c) Security Laws

Security laws are legal frameworks and regulations designed to protect information systems, networks, and digital data from misuse, unauthorized access, and cyber threats. They form the **legal backbone of cybersecurity**, ensuring accountability for cybercrimes and data breaches.

Key Objectives of Security Laws:

1. Define punishable cyber offenses and their penalties.
2. Regulate data handling, privacy protection, and information security.
3. Provide legal guidelines for digital evidence admissibility in court.
4. Establish frameworks for national and international cyber defense cooperation.

Examples of Security Laws:

1. **Information Technology Act (India, 2000):** Defines and penalizes cybercrimes such as hacking, data theft, identity fraud, and cyber terrorism.
 - *Section 43:* Addresses unauthorized access to computer systems.
 - *Section 66:* Penalizes hacking and data alteration.
2. **GDPR (General Data Protection Regulation – EU):** Protects personal data and privacy of EU citizens, regulating data collection and processing.
3. **HIPAA (Health Insurance Portability and Accountability Act – USA):** Secures electronic health records and medical information.
4. **Cybercrime Prevention Act (Philippines):** Establishes cybercrime definitions and penalties for online offenses.

Importance:

- Establishes accountability for digital misconduct.
- Safeguards user data and privacy.
- Promotes trust in online transactions and e-commerce.
- Supports law enforcement in investigating and prosecuting cybercrimes.

Example: Under Section 66C of India's IT Act, a cybercriminal involved in identity theft using fake social media accounts is prosecuted.

Q.12 Rewrite about the following: (Unit 1)

a) Internet Crimes

b) Credit Card Frauds

c) Emerging Digital Crimes

a) Internet Crimes

Internet crimes, also known as **cybercrimes**, are unlawful acts committed using the internet, computers, or digital communication networks as the primary medium. These crimes exploit vulnerabilities in online systems and target individuals, organizations, or governments for financial gain, data theft, disruption, or defamation.

Common Types of Internet Crimes:

1. **Phishing:** Deceptive emails or websites are used to steal login credentials and financial data.

2. **Identity Theft:** Attackers impersonate victims by stealing personal data such as Aadhaar, PAN, or credit card information.
3. **Hacking:** Unauthorized access to computer systems or networks to modify, steal, or destroy information.
4. **Cyberstalking:** Continuous online harassment, threats, or monitoring of a person's online activity.
5. **Ransomware Attacks:** Malware encrypts a victim's data, demanding payment to unlock it.

Impact of Internet Crimes:

- Loss of financial and personal data.
- Damage to organizational reputation.
- Breach of national security and privacy.

Example: A phishing email posing as a bank alert tricks users into revealing their account passwords, leading to unauthorized fund transfers.

Preventive Measures:

- Strong passwords and multi-factor authentication.
- Awareness and cybersecurity training.
- Use of firewalls, antivirus software, and secure network protocols (HTTPS, VPN).

b) Credit Card Frauds

Credit card fraud is a form of financial cybercrime where unauthorized individuals steal credit card information to conduct illegal transactions. These crimes often involve online payment systems, e-commerce platforms, or physical card theft.

Types of Credit Card Frauds:

1. **Card Skimming:** Cloning card details using a device installed on ATMs or POS machines.
2. **Phishing and Vishing:** Fraudulent emails or calls trick users into sharing card details.
3. **Online Payment Fraud:** Hackers exploit weak payment gateways to access sensitive card data.
4. **Data Breach Theft:** Hackers steal bulk credit card information from online databases.

Consequences:

- Financial loss for victims and banks.
- Damage to customer trust in online transactions.
- Legal and reputational risks for financial institutions.

Preventive Measures:

- Use of EMV chip-enabled cards.

- OTP (One-Time Password) verification for online payments.
- Regular monitoring of account statements.
- Implementation of **PCI DSS (Payment Card Industry Data Security Standard)** by banks and merchants.

Example: A hacker gains access to a shopping website's payment database and steals credit card details of thousands of users for fraudulent purchases.

c) Emerging Digital Crimes

Emerging digital crimes are the latest forms of cyber offenses that have evolved with technological advancements. These crimes exploit modern platforms such as artificial intelligence (AI), the Internet of Things (IoT), blockchain, and social media.

Types of Emerging Digital Crimes:

1. **Deepfake Manipulation:** Use of AI to create fake videos or images for defamation or blackmail.
2. **Cryptocurrency Frauds:** Scams involving fake crypto exchanges, wallets, or investment schemes.
3. **IoT Attacks:** Exploiting vulnerabilities in smart home devices like cameras and voice assistants.
4. **Cyber Espionage:** State-sponsored digital spying to steal confidential national or corporate data.
5. **Ransomware and Extortionware:** Attackers encrypt data and demand ransom, often threatening to release sensitive information publicly.

Impact:

- Financial losses running into millions.
- Privacy breaches and manipulation of public opinion.
- Threats to national security and personal safety.

Preventive Measures:

- Use of AI-based threat detection systems.
- Implementation of cyber awareness programs.
- Regular software updates and endpoint protection.
- Strong regulatory frameworks for cryptocurrency and data protection.

Example: A cybercriminal uses a deepfake video to impersonate a company's CEO and trick an employee into transferring funds to a fraudulent account.
